Visual Cryptography for Gray-level Image by Random Grids*

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Abstract

Visual cryptography is a kind of image encryption technology. It is characterized by the process of decryption that is overlapping the encrypted images and restores the secret by the human vision. Kafri and Keren proposed three random grid algorithms to encrypt a binary secret image in 1987. The advantage of their research is that the encrypted images with random grid will not be excessive expansion. This work modifies Kafri and Keren's random grid algorithm and proposes a new algorithm to encrypt gray-level secret images in two ways. As the result, this work greatly improves not only the color information in encrypted images but also the usability.

1 Introduction

With the progress of science and technology, the development of internet is getting much faster. In order to prevent the secret image not be stolen when transmits through the internet, the secret image must be encrypted, which is called *image encryption*. Many proposed image encryption researches [12, 13, 14, 15] used the random grid [1, 3, 4, 5, 6, 7, 8] algorithm to encrypt the secret image. By using this method, the size of generated image is as same as the original secret image, and the need for computing is also eliminated. Just when overlap those two encrypted images, the hidden secret of the original image can be shown.

In 1987, Kafri and Keren defined that each pixel of the random grid could only be classified into transparent (white) or opaque (black) [10]. And each pixel is generated by the random number, so the amount of the transparent pixels will equal to the opaque pixels. Therefore, the average light transmission ($\Im$) of the random grid is 1/2. Kafri et al. also proposed three different random grid algorithms to encrypt a binary secret image. The input of the algorithm is an image A and the outputs are two images R_1 and R_2 which are made by random grids. These three algorithms could only be used to encrypt a binary secret image. Hence, this paper will propose two methods and a new algorithm to encrypt a gray-level secret image that can output gray-level encrypted images.

This paper discusses the methods of gray-level encryption with random grid. The proposed scheme classifies all pixels into more than two colors to encrypt a gray-level secret image. The input of the new algorithm is a gray-level secret image B and the outputs are two gray-level encrypted images G_1 and G_2 which are made by random grids. When the two encrypted images are superposed, user can see the hidden secrets through the gray-level image. The purpose of this study is to improve the color information in encrypted images. It is different from Kafri et al.'s method which only classifies all pixels into two colors to encrypt a binary secret image. This work can make the color information more plentiful and greatly enhance the usability.

In the rest of this paper, Section 2 indicates the principle of gray-level encryption with random grids and provides two methods (ways) to encrypt every pixel on the gray-level secret image. Section 3 illustrates the experimental result of this study. Based on the Section 3, Section 4 will have some discussions and conclusions about this work.

2 The Proposed Scheme

Kafri et al. proposed three different random grid algorithms to encrypt a binary secret image. The input of the algorithm is a $w \times h$ image, denoted by A, and the outputs are two images R_1 and R_2. One of their algorithms is shown as below.

Algorithm 1:

Generate a $w \times h$ random grid R_1// $\Im(R_1) = 1/2$
for($i = 0$; $i < w$; i ++)
  for($j = 0$; $j < h$; j ++)
    if($A[i][j] == 0$)
      $R_2[i][j] = R_1[i][j]$;
    else
      $R_2[i][j] = \overline{R_1[i][j]}$;
output (R_1 , R_2)

Based on the above algorithm, this work proposes a new algorithm which is designed to

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process one gray-level secret image, denoted by B, and generates two gray-level encrypted images, denoted by G_1 and G_2, that all pixels are classifed into more than two colors. When user overlaps those two encrypted images G_1 and G_2, the hidden secrets of the gray-level image B can be shown. According to the range of RGB value in gray-level, two methods below are conclouded to encrypt every pixel on the gray-level secret image.

<u>Method 1:</u>

As the instance, given a gray-level secret image B whose pixels can be classified into 5 colors, where the RGB value of each can be thought as arithmetic progression. By referring to the color of every pixel in image B, this method randomly chooses the color combination for pixels, which are on the same position, in G_1 and G_2. TABLE 1 shows all the color combination probabilities of pixels in G_1 and G_2, and the relation with B. Each "4", "3", "2", "1" and "0" in TABLE 1 represents the RGB value (0, 0, 0), (64, 64, 64), (128, 128, 128), (192, 192, 192) and (255, 255, 255), respectively.

TABLE 1

B	G_1	G_2
0	0	0
0	1	1
0	2	2
0	3	3
0	4	4
1	0	1
1	1	0
1	2	2
1	3	3
1	4	4
2	0	2
2	1	1
2	2	0
2	3	3
2	4	4
3	0	3
3	1	2
3	2	1
3	3	0
3	4	4
4	0	4
4	1	3
4	2	2
4	3	1
4	4	0

Referred to the above example, when a gray-level secret image B whose pixels can be classified into 23 colors, we can easily speculate the combination probabilities of G_1 and G_2 in the similar way. In TABLE 2, the RGB value of "23", "22", "21", ..., and "0" are also in arithmetic progression, which represents the RGB value (0, 0, 0), (10, 10, 10), (21, 21, 21), ..., (255, 255, 255), respectively.

TABLE 2

B	G_1	G_2
0	0	0
0	1	1
0	2	2
$\vdots$	$\vdots$	$\vdots$
0	20	20
0	21	21
0	22	22
$\vdots$	$\vdots$	$\vdots$
i	0	i
i	1	$i-1$
$\vdots$	$\vdots$	$\vdots$
i	$i-1$	1
i	i	0
i	$i+1$	$i+1$
i	$i+2$	$i+2$
$\vdots$	$\vdots$	$\vdots$
i	21	21
i	22	22
$\vdots$	$\vdots$	$\vdots$
22	0	22
22	1	21
22	2	20
$\vdots$	$\vdots$	$\vdots$
22	20	2
22	21	1
22	22	0

Now, we can conclude a general algrithm from above two examples. When we classify the pixel into n colors, where $n \geq 2$, we can use the general algrithm as below.

<u>Algorithm 2:</u>

Generate a $w \times h$ random grid G_1// $\Im(G_1) = 1/2$
for($i = 0$; $i < w$; i ++)
    for($j = 0$; $j < h$; j ++)
        if($B[i][j] < G_1[i][j]$)
            $G_2[i][j] = G_1[i][j]$;
        else

$$G_2[i][j] = B[i][j] - G_1[i][j];$$

Note that Algorithm 1 will be a special case of Algorithm 2 when $n = 2$.

Method 2:

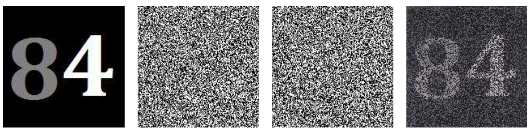
This method provides another idea to classify the color of gray-level pixels. The differences between each RGB value of gray-level pixels are similar to geometric progression. The principle of Method 2 is as same as Method 1, such as the process of encryption and the algorithm. The only different place is the RGB value of each gray-level pixel. For example, different from TABLE 1, each "4," "3," "2," "1" and "0" represents the RGB value (0, 0, 0), (25, 25, 25), (76, 76, 76), (153, 153, 153) and (255, 255, 255), respectively.

3 Experimental Result

This paper proposes one new algorithm in two methods (ways) for gray-level encryption with random grids. This section presents the results of Method 1 and Method 2 which have mentioned in previous section. Subsequently, the followings are results of encrypting gray-level images using Method 1 and Method 2. The first 8 results adopt the former, and the rest adopt the latter, respectively.

Method 1:

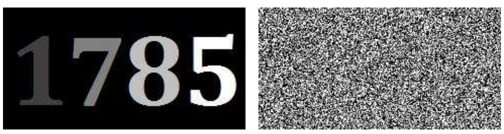
Result 1: All pixels of a gray-level number secret image B are classified into 3 colors which is shown in Figure 1(a). Figures 1(b) and 1(c) indicate the two gray-level encrypted images G_1 and G_2 and Figure 1(d) is the decrypted image which is produced by overlapping G_1 and G_2.



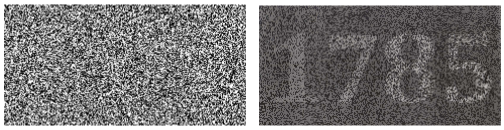
(a) (b) (c) (d)

Fig. 1 Result 1: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

Likewise, Result 2, 3 and 4 classify all pixels into 5, 11 and 23 colors, respectively. The experimental results are shown in Figure 2, 3 and 4.

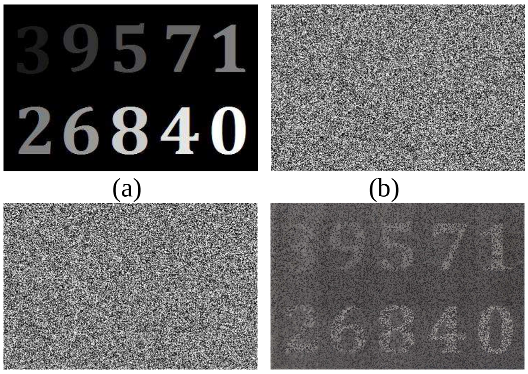


(a) (b)



(c) (d)

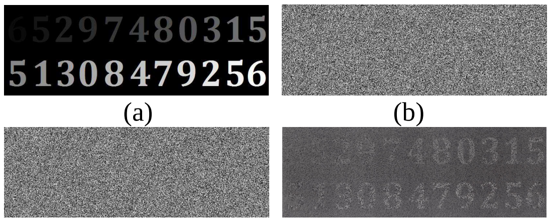
Fig. 2 Result 2: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.



(a) (b)

(c) (d)

Fig. 3 Result 3: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

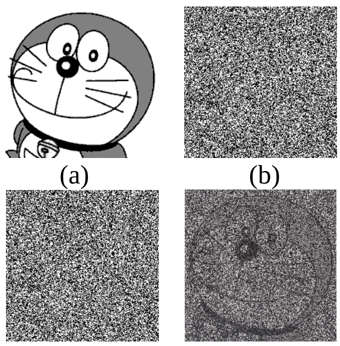


(a) (b)

(c) (d)

Fig. 4 Result 4: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

Result 5: Each pixel is classified into 3 colors, but the input is not a gray-level secret image that only with ciphers on it. The result is shown in Figure 5.

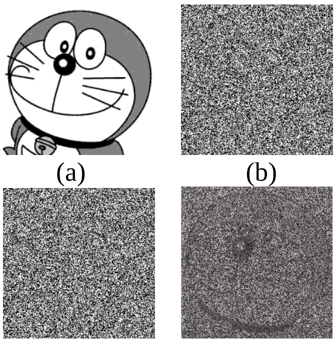


(a) (b)

(c) (d)

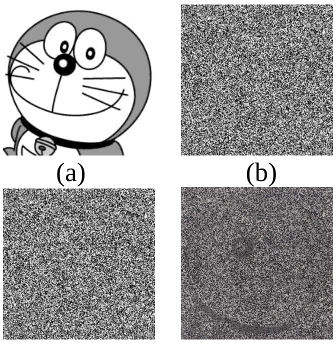
Fig. 5 Result 5: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

Similarily, Result 6, 7 and 8 classify all pixels into 5, 11 and 23 colors, respectively. Figure 6, 7 and 8 indicate the results.



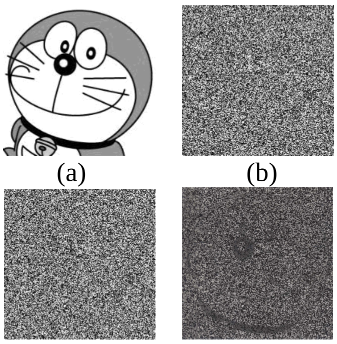
(a) (b) (c) (d)

Fig. 6 Result 6: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.



(a) (b) (c) (d)

Fig. 7 Result 7: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

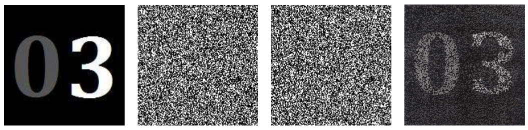


(a) (b) (c) (d)

Fig. 8 Result 8: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

Method 2:

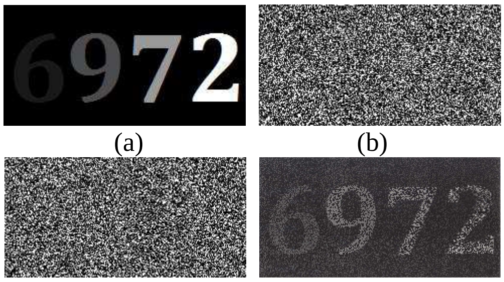
Result 9: Given a gray-level number secret image B whose pixels can be classified into 3 colors that is shown in Figure 9(a). Figures 9(b) and 9(c) show the two gray-level encrypted image G_1 and G_2 that obtained by Algorithm 2. Figure 9(d) illustrates the decrypted image which is generated by overlapping G_1 and G_2.



(a) (b) (c) (d)

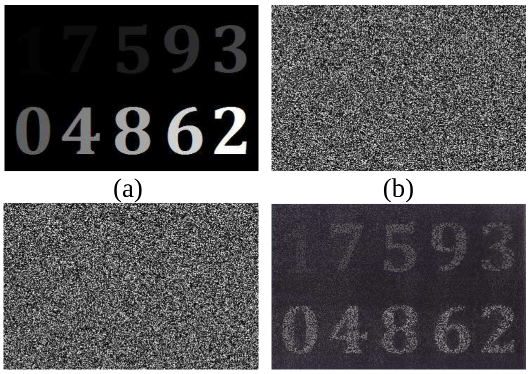
Fig. 9 Result 9: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

Result 10, 11 and 12 are similar to Result 9, but each of them uses the pixels which could be classified into 5, 11 and 23 colors, respectively. The results are shown in Figure 10, 11 and 12.



(a) (b) (c) (d)

Fig. 10 Result 10: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.



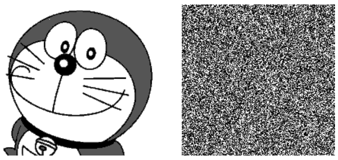
(a) (b) (c) (d)

Fig. 11 Result 11: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

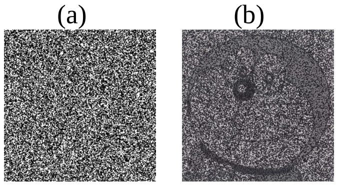
(a) (b) (c) (d)

Fig. 12 Result 12: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

Result 13: In this experiment, the image with not only ciphers is applied as the input. All pixels classify into 3 colors are used. The result is shown in Figure 13.



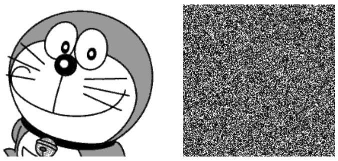
(a) (b)



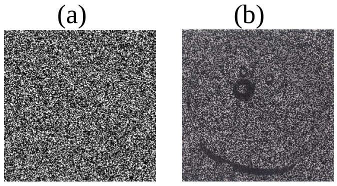
(c) (d)

Fig. 13 Result 13: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

In the same way, Result 14, 15 and 16 consist of the pixels which could be classfied into 5, 11 and 23 colors, respectively. Figure 14, 15 and 16 display the results mentioned above.

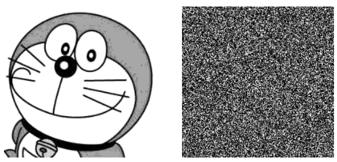


(a) (b)

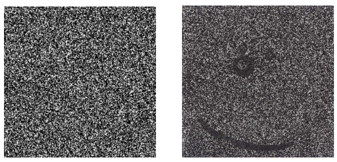


(c) (d)

Fig. 14 Result 14: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

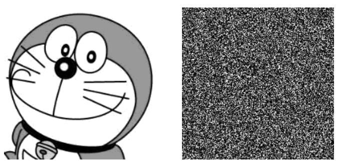


(a) (b)

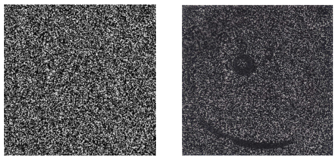


(c) (d)

Fig. 15 Result 15: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.



(a) (b)



(c) (d)

Fig. 16 Result 16: (a) the secret image B, (b) the encrypted image G_1, (c) the encrypted image G_2, (d) the decrypted image.

4 Discussion and Conclusion

According to previous experimental results, this section compares two methods proposed in this work and has some discussions.

At first, we need to explain how we got the decrypted images (d) of Figure 1-16 in previous section. Actually, we print the encrypted images G_1 (b) and G_2 (c) into two transparencies for each case, then scan the result image by overlapping these two transparencies to obtain the image which shows as (d) in each Figure 1-16. Hence, there may some residuals cause by operant. Please keep it in mind.

This work finds out that user can see the decrypted image which uses Method 1 more clearly than which uses Method 2, when the input is not a gray-level secret image which has ciphers only. On the contrary, when the input is a gray-level secret image with only ciphers on it, the decrypted image is clearer by using Method 2. Moreover, the disadvantage of Method 2 is that the pixels could only be classified into 23 colors at most. It is different from the Method 1 which could classify the pixels into 256 colors maximum.

The proposed reserch uses two ways to encrypt a gray-level secret image. One of them is good at encrypting pitures, and the other is adept in encrypting numbers or texts. Alternatively, the proposed methods could be choosen on demand to encrypt a gray-level image, that make the color information more plentiful and also improve the limits of those methods adopting 2 colors.

In fact, there exist some related works that discuss about encrypt a gray-level secret image or color image [2, 9, 11, 13]. Here, we list the dissimilarities between them and this paper. In 1995, Naor and Shmair [13] proposed a visual cryptography for gray-level Image, called NS scheme, the main idea is to extend each pixel serval times and use codebook to encrypt the secret image. In 2000, Blundo et al.[2] gave a necessary and sufficient condition for the schemes in [13] to exist. They show that NS scheme should have at least $(g - 1)*2^{k-1}$ expansions, where g is grey levels and k is the number of encrypted images. For example, if g = 11 and k = 2 , the encrypted images are 20 times than the secret image. After that, Lin is also proposed another visual cryptography for gray-level Image in 2003 [11]. Their scheme has a preprecessing by dithering techniques, and then tranfer the gray level image to a binary image. Hence, The kernel of their scheme is still to encrypt a binary secret image by extending pixel size and use codebook. Beside, visual cryptography for color image is also proposed by [9] in 2003. In [9], Hou divides a color image to RGB three images, and still tranfer these images to a binary images and encrypt a binary secret image by extending pixel size and use codebook. Hence, above serval visual cryptography for gray-level or color image are different with our scheme.

References

[1] J.-L. Bai, *Random-based secret image sharing scheme*, Master's Thesis, Computer Science and Information Engineering, Ming Chuan University, 2005.

[2] C. Blundo, A. D. Santis, M. Naor: *Visual cryptography for grey level images.* Information Process. Letter, vol.75, no.6, pp. 255-259, 2000.

[3] T.-H. Chen, G.-Z. Wei, and K.-X. Taso, "An multi-secret image scheme by using random grids," *in Proceedings of 18th Information Security Conference*, Hualien, May 29-30, 2008.

[4] T.-H. Chen, K.-H. Tsao, and K.-C. Wei, "Multiple-image encryption by rotating random grids," *in Proceedings of The 8th International Conference on Intelligent System Design and Applications (ISDA 2008)*, vol. 3, 2008, pp. 252-256.

[5] T.-H. Chen, and K.-H. Tsao, "Image encryption by (n, n) Random Grids," *in Proceedings of 18th Information Security Conference*, Hualien, May 29-30, 2008.

[6] T.-H. Chen, K.-H. Taso, and Y.-T. Yang, "Friendly color visual secret sharing by random grids," *Fundamenta Informaticae*, vol. 96, 2009, pp. 61-70.

[7] Y.-Y. Chen, Justie S.-T. Juan, and M.-J. Li, "A random grid-based visual cryptography to adjust light transmission and the number of shares," *in Proceeding of 2009 National Computer Symposium*, National Taipei University, Taiwan, Nov. 27-28, 2009, pp. 184-192.

[8] Joy Jo-Yi Chang, Ming-Jheng Li, Yi-Chun Wang and Justie Su-Tzu Juan, "Two-Image Encryption by Random Grids," International Symposium on Communications and Information Technologies (ISCIT), Oct. 2010, pp. 458-463.

[9] Y.-C. Hou, *Visual cryptography for color images*. Pattern Recognition, vol.36, no.7, pp.1619-1629, 2003.

[10] O. Kafri, and E. Keren, "Encryption of pictures and shapes by random grids," *Optics Letters*, vol. 12, no. 6, 1987, pp. 377-379.

[11] C.-C. Lin, W.-H. Tsai: Visual cryptography for gray-level images by dithering techniques. Pattern Recognition Letters, vol.24, no.1-3, pp.349-358, 2003.

[12] R. Lukac, and K. N. Plataniotis, "Bit-level based secret sharing for image encryption," *Pattern Recognition*, vol. 38, no. 5, 2005, pp. 767-772.

[13] M. Naor, and A. Shamir, "Visual cryptography," *in Proceedings of Advances in Cryptology – Eurocrypt' 94*, Lecture Notes in Computer Science, vol. 950, Springer Berlin, 1995, pp. 1-12.

[14] S.-J. Shyu, "Image encryption by random grids," *Pattern Recognition*, vol. 40, no. 3, 2007, pp. 1014-1031.

[15] S.-J. Shyu, "Image encryption by multiple random grids," *Pattern Recognition*, vol. 42, no. 7, 2009, pp. 1582-1596.